ENBC332: Homework #6

Due: Monday, 11/03/25 by noon, beginning of the class

Submit your homework to ELMS, including R code and dataset

Overall points: 40

Note: Late homework may be submitted up to 24 hours after the due date, but a 50% penalty will be applied.

Note: Please aim to write clean, well-organized code that is easy for others to understand. Be sure to include comments or explanations where needed. Points may be deducted if this is not done.

Note: Please name your codes using the following format: "problem1_fullname.R". Start with the problem number and followed by your full name, e.g., "**problem1_jimmyazarnoosh**".

Note: Using generative AI for learning purposes is allowed. However, directly copying or closely replicating AI-generated content is strictly prohibited and will result in a grade of **ZERO**. Repeated violations may lead to more severe consequences.

Note: Submit **a single zip file** containing the following:

- The dataset you used
- R code files
- Your answers to the questions in PDF format

Note: Functions are recommended to follow the format below:
e.g., calculating range.

```
# --------------------------------------------------
Data <- trees$Height
my_range_func <- function(Data){
  Range <- max(Data) - min(Data)
  return(Range)
}
# --------------------------------------------------
```

P-value:

Problem 1. This problem helps you understand how to conduct a one-sample t-test in R, interpret the resulting p-value, and make decisions based on hypothesis testing. (20 points)

A nutritionist claims that the average daily sodium intake for adults in a specific population is 2,300 mg. To test this claim, you measure the daily sodium intake of 30 randomly selected adults and get the following values (in mg):

- **Sodium Intake Data (in mg):** 2100, 2500, 2300, 1950, 2600, 2200, 2450, 2150, 2400, 2550, 2350, 2250, 2050, 2650, 2300, 2400, 2100, 2500, 2350, 2200, 2450, 2150, 2250, 2550, 2050, 2600, 2300, 2150, 2400, 2500

Write R Code: (10 points)

- Use R to perform a one-sample t-test.
- Calculate the p-value and explain its meaning in the context of this problem. Use a significance level of 0.05.

Conceptual questions:

1. State the null hypothesis (H_0) and the alternative hypothesis (H_1) to test if the average daily sodium intake is significantly different from 2,300 mg. (4 points)
2. Describe what the p-value tells you about the average sodium intake compared to the population mean. (3 points)
3. State whether you reject or fail to reject the null hypothesis and explain the practical implication of this decision. (3 points)

Problem 2. This problem helps you understand how to conduct a one-sample t-test in R, interpret the resulting p-value, and make decisions based on hypothesis testing. (20 points)

A pharmaceutical company claims that a new cholesterol-lowering drug reduces the average LDL cholesterol level in adults to less than 120 mg/dL after 8 weeks of treatment. To test this claim, you measure the LDL cholesterol levels of 25 randomly selected adults who took the drug and get the following values (in mg/dL):

- **LDL Cholesterol Data (in mg/dL):** 110, 112, 115, 113, 111, 114, 109, 112, 111, 115, 113, 110, 112, 109, 111, 113, 114, 110, 112, 111, 109, 113, 112, 110, 111

Write R Code: (10 points)

- Use R to perform a one-sample t-test.
- Calculate the p-value and explain its meaning in the context of this problem. Use a significance level of 0.05.

Conceptual questions:

1. State the null hypothesis (H_0) and the alternative hypothesis (H_1) to test if the average LDL cholesterol is less than 120 mg/dL. (4 points)
2. Describe what the p-value tells you about the average LDL cholesterol compared to the target value. (3 points)
3. State whether you reject or fail to reject the null hypothesis and explain the practical implication of this decision. (3 points)